



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Falcon Landing; D.R. Horton Atlanta West
Location: Gainesville, GA
Description: Belford - D2/D3, Slab

Date: September 22, 2024
By: TJH
House Faces: Worst East

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,248 ft ²
Volume	20,049 ft ³
Percentage Glass	11 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	8851
SEER	15
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x6	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.5	Exterior Door		R-2	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling	2x4	R-38 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.35	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		27067 Btu/hr		31720 Btu/hr	
Minimum Nominal Capacity		3.01 Tons		9.29 kW	
Load Density		747 SF/Ton		4.1 W/SF	
Floor Area Served		2248.0 SF		2248.0 SF	
Mechanical Ventilation		N/A CFM		N/A CFM	
System Size		3.0 Tons		15.0 kW	
Leaving Air Temperature		60.1 deg F		103.9 deg F	
Installed Capacity		749 SF/Ton		6.7 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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[illegible]

Unit 1

Size

3.00

Tons

Airflow

1200

CFM

Heat kW

15

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer & Closet	32	56	44	
Hall	9	12	10	
Powder	3	6	4	
Pantry	2	3	3	
Bed 1 WIC	10	21	15	
Bath 1	29	52	41	
Bed 1	125	140	133	
Great Rm	251	235	243	
Laundry	41	19	30	
Garage Entry Hall	24	38	31	
Kitchen	154	55	104	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	67	122	94	
Landing / Hall	51	55	53	
Bed 4 WIC	6	9	8	
Bed 4	69	67	68	
Upstairs Living	113	49	81	
Bed 3	74	84	79	
Bath 2	25	39	32	
Bed 2 WIC	25	39	32	
Bed 2	90	101	95	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

DX Cooling kWh	1911
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RC Heating kWh	2904
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Total kWh	4815
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SEER 15

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Cooling Size	1.5	DX Cooling kWh	1458
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By: TJH

Heating Size	1.5	RC Heating kWh	2579
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Level(s): Second Floor

SEER 15

Cooling Load Calculation							Ceiling Height		18		8		8		8		8		8		8		8		8		8		8		8									
Components							CLTD (°F)		U-Value (Btu/hr-sq-F)		Cooling Load Factor		Entire House		Stairs		Landing / Hall		Bed 4 WIC		Bed 4		Upstairs Living		Bed 3		Bath 2		Bed 2 WIC		Bed 2									
	Type	Description					Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr		
Glass	G1	North	Right	(E)	6	Btu/hr-sf	15	93		-		-		-		-		-	15	93		-		-		-		-		-		-		-		-		-		
	G1	NorthEast		(SE)	8	Btu/hr-sf	0	0																																
	G1	East	Front	(S)	13	Btu/hr-sf	90	1175	30	392	15	196												15	196	30	392													
	G1	SouthEast		(SW)	14	Btu/hr-sf	0	0																																
	G1	South	Left	(W)	16	Btu/hr-sf	15	237						15	237																									
	G1	SouthWest		(NW)	26	Btu/hr-sf	0	0																																
	G1	West	Back	(N)	29	Btu/hr-sf	0	0																																
	G1	NorthWest		(NE)	18	Btu/hr-sf	0	0																																
	G1	Sky Light			46	Btu/hr-sf	0	0																																
G1	Shaded			6	Btu/hr-sf	15	93	15	93																															
Door	D1	Door		21	0.50	10.50	Btu/hr-sf	0	0																															
	D2	Door		55	0.50	27.50	Btu/hr-sf	9	256						9.32	256																								
Above Grade Walls	W1	North		13	0.08	1.00	Btu/hr-sf	273	273									114	114	106	107	52	53																	
	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	0	0																															
	W1	East		23	0.08	1.77	Btu/hr-sf	347	615	164	290	48	85											41	72	94	167													
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0																															
	W1	South		16	0.08	1.23	Btu/hr-sf	351	433	149	183			33	41	98	120									72	89													
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0																															
	W1	West		23	0.08	1.77	Btu/hr-sf	113	199							52	92																							
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	0	0																															
Basement Walls	W1	North		13	0.08	1.00	Btu/hr-sf	0	0																															
	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	0	0																															
	W1	East		23	0.08	1.77	Btu/hr-sf	0	0																															
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0																															
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0																															
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0																															
	W1	West		23	0.08	1.77	Btu/hr-sf	0	0																															
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	0	0																															
Other	W2	Grade		0	0.07	0.00	Btu/hr-sf	0	0																															
	W1	Partition		10	0.08	0.77	Btu/hr-sf	0	0																															
	W1	Garage		30	0.08	2.31	Btu/hr-sf	0	0																															
	W3	Attic		55	0.05	2.92	Btu/hr-sf	220	642						52	151	108	314	61	177																				
	W3	Kneewall		55	0.05	2.92	Btu/hr-sf	0	0																															
Floor / Roof	R1	Ceiling		55	0.03	1.43	Btu/hr-sf	1043	1491	86	123	163	233	23	33	144	206	162	232	185	265	73	104	36	51	171	245													
	R2	Vaulted		55	0.05	2.92	Btu/hr-sf	0	0																															
	F1	Crawl/Uncon.		11	0.05	0.58	Btu/hr-sf	0	0																															
	F1	Garage		30	0.05	1.59	Btu/hr-sf	409	650							5	8	41	65	73	116	36	57	171	272															
	F2	Slab		0	0.10	0.00	Btu/hr-sf	60	0	60	0																													
Internal	People				255	Btu/hr	4	1020						1	255	1	255	1	255						1	255														
	Lights (W)				3.413	Btu/hr-W	524	1788	43	147	82	280	12	41	72	246	81	276	93	317	37	126	18	61	86	294														
	Appliances (W)				3.413	Btu/hr-W	250	853									250	853																						
Vent	Infiltration		0.25	ACH	0.09	Btu/hr-cf		828			139	117		17		104		117		133		53		26		123														
	Volume Area				CF SF		9204 1043		1548 86	1304 163	184 23	1152 144		1296 162	1480 185	584 73		288 36		1368 171					0		0		0		0		0		0					
Subtotal (Btu/hr)								10648		1368	1043	131	1410		2311	1527	506	516	1836		0		0		0		0		0		0		0		0					
Safety Factor								1065																																
Sensible Cooling Load (Btu/hr)								11713																																
Nominal Tons Required (75% sens. cap.)								1.30																																
System Selection 1.5 Tons Space CFM								600		77	59	7	79	130	86	29	29	103		-		-		-		-		-		-		-		-		-				
HEATING LOAD CALCULATION																																								
Components							Heating Load		Entire House		Stairs		Landing / Hall		Bed 4 WIC		Bed 4		Upstairs		Bed 3		Bath 2		Bed 2 WIC		Bed 2		0		0		0		0					
	Type	Description	ΔT (°F)	U	Val	g	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr				
Glass	G1		54	0.35	18.9	Btu/hr-sf	135	2552			45	851	15	284	0	-	15	284	0	-	15	284	0	-	15	284	30	567	0	-	0	-	0	-	0	-				
	G2		54	0.00	0.0	Btu/hr-sf	0	0			0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
	G3		54	0.00	0.0	Btu/hr-sf	0	0			0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
Door	D1		54	0.50	27.0	Btu/hr-sf	0	0			0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
	D2		54	0.50	27.0	Btu/hr-sf	9	252			0	-	0	-	0	-	9.32	252	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
Walls	W1		54	0.08	4.2	Btu/hr-sf	1084	4506	313	1,300	48	200	33	137	149	621	0	-	175	726	106	443	93	387	166	692	0	-	0	-	0	-	0	-	0	-				
	W2		54	0.07	3.6	Btu/hr-sf	0	0			0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
	W3		54	0.05	2.9	Btu/hr-sf	220	630			0	-	0	-	51.8	148	108	308	60.8	174	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
	W1	Partition	27	0.08	2.1	Btu/hr-sf	0	0			0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
	W2	Grade	27	0.07	1.8	Btu/hr-sf	0	0			0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-				
Floor / Roof	R1		54	0.03	1.4	Btu/hr-sf	1043	1464	86	121	163	229	23	32	144	202																								